

A Project Report on

Simulink based Embedded Coding Regen Cum Start-Stop using Ultra (Super) Capacitor

Submitted by
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In partial fulfillment for the award of the degree of
Bachelor of Technology
In
Electronics and Communication Engineering



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BONAFIDE CERTIFICATE

Certified that the project work entitled “**Simulink based Embedded Coding - Regen Cum Start-Stop using Ultra (Super) Capacitor** ” submitted to SASTRA University, Thanjavur by SWETHA.V (Reg.No:113004204) in partial fulfillment for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering is the work carried out independently under my guidance during the period Dec 2012 – April 2013.

Project Guide

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DECLARATION

I submit this project work entitled **“Simulink based Embedded Coding - Regen cum Start-Stop using Ultra (Super) Capacitor”** to SASTRA University, Thanjavur in partial fulfillment of the requirements for the award of the degree of **“Bachelor of Technology”** in **“Electronics and Communication Engineering”**. I declare that it was carried out independently by me under Prof J.Siva, AP III, ECE, SASTRA University as Internal Guide and Mr. V Narayanan, General Manager, Advanced Engineering, Lucas-TVS, Padi, Chennai as External guide.

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CHAPTER 1

INTRODUCTION

1.1 MOTIVATIONAL OVERVIEW

The number and importance of embedded systems has increased dramatically over the last few years. In this process, the complexity and criticality of these systems has reached a level that has become difficult to handle with classical development methods. As a consequence, the industry is currently in the process of switching to a new graphical model-based development approach. This approach is promising considerable gains in productivity through an increased level of abstraction and through the support for simulation and code-generation offered by the corresponding tools. The real challenge to the embedded system designers is to create a system which has complete thoroughness, easy understandability and quicker system design. Dr. Thierry Lestable, Samsung UK on their successful Development of 4G Wireless Systems with Simulink says - "Simulink enables us to easily share proposals and knowledge with other design centers. Simulink also allows us to focus on algorithm design and perform state-of-art mathematical analysis, assessment, simulation, and optimization." Owing to these advantages and out-standing performance achievable through Simulink, effort has been made to develop the control logic for the energy management in automobiles. Environmental pollution, Global warming and Energy constraints are the serious issue facing the world today. The power cuts for several hours a day is a serious issue faced by India and several other countries. It has become mandatory to conserve energy wherever, whenever possible. As an engineer, the product we develop for the public must

mainly focus on these issues. Regenerative braking and auto start-stop techniques are most promising technologies to combat these issues in a more effective way in the automobiles.

1.2 OBJECTIVE

The objective of this project work is three fold.

- Exploiting the Simulink- Stateflow approach as a replacement to the conventional coding. Comparing its merits with that of the conventional coding by weighing various parameters.
- Combining both the start-stop and regenerative braking concept as a means to circumvent the long-lasting energy crisis problem. Appreciation is given to the energy recovery that can be obtained by this dual technology.
- Exploring the Ultra-capacitor concept and combining it with the dual technology as mentioned above.

The software logic's approach is given more importance in this project work. The simulation results and the laboratory testing have been performed to verify the proposed concept.

1.3 PROJECT OUTLINE

The remaining part of the thesis is outlined as follows. Chapter 2 deals with the concept of embedded coding. It highlights the importance of graphical programming and a comparison with traditional coding has been drawn. Chapter 3 deals with the State-flow approach. It says about the difference between the conventional flow chart and state-flow. It explains how a system can completely be defined in a state-flow approach. Chapter 4 discusses the start-stop concept. It gives the knowledge on the recent trends, advantage, limitations of the auto-stop and also the

working of the proposed concept. Chapter 5 involves the concept of Ultra-capacitors. It discusses the difference between normal capacitor and ultra-capacitor. It outlines the use of ultra-capacitor in the energy saving systems. Chapter 6 describes the Regenerative braking concept. It discusses the proposed concept and its working. It also tells about the recent trends and challenges faced in this technology. Chapter 7 lists the features of the controller used in this project. It briefly describes the feature that is used pertaining to the project at hand. Chapter 8 outlines the hardware architecture used in the implementation of the above mentioned concept. Chapter 9 describes the software architecture behind this project work. It details the models used in the Simulink environment for the generation of code. Chapter 10 discusses the results obtained from the laboratory testing. The thesis concludes with Chapter 11 briefing the limitations and future scope of the project.

CHAPTER 2

Embedded coding through Simulink:

2.1 Introduction:

Simulink, developed by Math Works, is a data flow graphical programming language tool for modeling, simulating and analyzing multidomain dynamic systems. Its primary interface is a graphical block diagramming tool and a customizable set of block libraries. Simulink is widely used in control theory and digital signal processing for multidomain simulation and Model-Based Design. The following are the key features of Simulink.

- Graphical editor for building and managing hierarchical block diagrams
- Libraries of predefined blocks for modeling continuous-time and discrete-time systems
- Simulation engine with fixed-step and variable-step ODE solvers
- Scopes and data displays for viewing simulation results
- Project and data management tools for managing model files and data
- Model analysis tools for refining model architecture and increasing simulation speed
- MATLAB Function block for importing MATLAB algorithms into models
- Legacy Code Tool for importing C and C++ code into models

2.2 Embedded coding:

Embedded Coder is one of the unique features of Simulink. It generates readable, compact, and fast C and C++ code for use on embedded processors, on-target rapid prototyping boards, and microprocessors used in mass production. Embedded Coder enables additional MATLAB Code

and Simulink Coder configuration options and advanced optimizations for fine-grain control of the generated code's functions, files, and data. These optimizations improve code efficiency and facilitate integration with legacy code, data types, and calibration parameters used in production. Using this coder, the graphical environment is converted into an equivalent C or C++ code in the respective IDE and executed in the target processor/controller. We are using this Simulink embedded coder in this project for the application of regenerative braking and stop-start control in automobiles. The following features enable us to understand the importance of embedded coding when compared with traditional coding.

2.3 Embedded coding versus traditional coding:

Embedded code generation is fundamentally changing the way engineers work. Instead of writing thousand lines of code by hand, engineers are automatically generating their production code to increase productivity, improve quality and foster innovation.

- **Easier understandability:**

The coding environment is done in a graphical interface. This kind of block-wise representation of the functionality enables the user to easily understand the motivation behind the developed code. One need not look carefully into the entire code as in the case of traditional coding.

- **Standard coding format:**

Since there is usually “more than one way to do it”, it is necessary to establish a common style within a large project team in order to achieve consistent appearance of the final product. When generating code from Simulink through embedded coder, the generated

code has a pre-defined format in which it should appear in the project file. This enables us to create a uniform final code.

- **Time efficient:**

Several hundred lines of code is equivalent to single block. This reduces the time that is required to write an embedded code specific to a particular controller/processor. Also, the way in which the interrupt routine and function call are written in the traditional coding is greatly made simpler with Simulink environment.

- **Simulation capabilities:**

After developing a traditional code, we generally run it with the actual target to verify the results. But, with embedded coding, we can create models which are equivalent to the actual target and vary the parameters in the Simulink environment and thereby validate the code before dumping into the controller.

- **Interfacing controller:**

With the help of the blocks, it is easier to determine the controller features and configure the parameters without even having a sound knowledge on the registers specific to the controller in hand.

- **Universal code:**

The same Simulink block can be used to generate code specific to a variety of controller or processor. This reduces the time to write code specific to different controllers/processors for the same application.

2.4 CODE GENERATION:

Embedded Coder supports code generation and verification on Texas Instruments C2000, C5000, C6000 and OMAP family of processors. The Code Composer Studio IDE is also supported during the code generation process through make file based build automation. With Embedded Coder, we can generate code from your Simulink models and implement it directly on supported processors, perform PIL tests, and gather execution performance data through the profiling feature. Evaluation boards listed below are also supported for full turnkey prototyping by providing libraries for on-chip and onboard peripherals and supporting code generation of real-time operating systems for select processor families. The project uses the TMS320F28035 controller with CCS as the IDE link. The supporting experimental kit is used to verify the generated code in the actual target environment. Simulation is done in Simulink environment.

2.5 RECENT TRENDS:

ARM and STMicroelectronics have announced the availability of the first ever Embedded Coder support, with MATLAB and Simulink, for ARM Cortex-M processor-based systems. The joint project between ST and ARM is fully supported by Math Works and enables software developers to create their algorithms in MATLAB and Simulink and then target, integrate, debug and test those models in a Processor in the Loop simulation.

The generated C code from Embedded Coder runs on an STM32 Evaluation Board and the debugger of the Keil Microcontroller Development Kit interfaces directly with Simulink, further simplifying the code integration.

CHAPTER 4

START STOP SYSTEM

4.1 Introduction:

A stop-start system automatically shuts down the engine when the vehicle comes to a stop and restarts the engine to reduce the amount of time a car spends idling. The vehicle employs a beefier starter that can withstand a significantly increased number of engine starts over the life of the vehicle. A more powerful and robust battery is crucial. A stop-start system requires a battery that is power dense and able to withstand multiple stop start events.

An engine already at optimum temperature, though, stopping and starting does not use an excessive amount of fuel. This is why manufacturers have introduced systems that automatically limit idling in a motionless vehicle. Volkswagen debuted its first production stop-start system in 1983 with the Europe-only Polo Formel E.

All these systems work similarly and are ubiquitous on hybrids. When the car comes to a stop, the engine computer cuts spark and fuel. When the driver lifts his foot off the brake, or engages the clutch, the engine fires back up.

With many drivers' minds on the environment and the increasing costs of oil production and consumption these days, people are looking for alternatives to the conventional gasoline engine. One way to combat the issue of frequently stopping, starting and standing still is a technology that can seamlessly switch your engine on and off depending on how your vehicle is operating.

An idle-stop system, also commonly known as a start-stop system, is essentially what makes up today's increasingly popular so-called mild hybrids.

4.2 Stop – start controller:

The brains behind engine stop-start systems are the computer programs that control them. But drivers don't interact with the engine computer, so for the person in the pilot's seat, it's the accelerator and brake pedal that make the magic happen.

Initially, when the ignition is turned ON, the LED blinks once confirming the proper functioning of the controller. The engine is started first using the key. The engine is assumed to be fired only when it reaches a particular frequency.

When there is a battery dip, auto-stop feature is not enabled. The auto-stop mode is enabled only when engine is idling and the vehicle speed had crossed the initial threshold. This is to avoid the confusion of the feature during initial start-up and facilitate it only during the necessary conditions. The auto-stop enable-disable switch is provided to the user as a friendly means to enable or disable this feature. Thus, this feature is enabled only when the user enables the switch. Even if the vehicle is push started, the auto-stop feature is enabled. The LED blinking is provided as a means to provide an indication that vehicle is going to be stopped due to this feature.

Stop-start systems are designed so that depressing the brake pedal sends a "ready" signal to the engine controller. The clutch pedal is often the control device on vehicles with manual transmissions.

When the car comes to a complete stop, the controller shuts the engine down and — depending on the design of the system and type of transmission — pre-positions the starter motor, transmission and fuel injection system to provide instantaneous engine restart when the driver presses the clutch pedal.

Safety features such as vehicle's bonnet, starter operating time are taken into consideration. Example, the starter can be damaged due to repeat engaging with the engine. Thus, overcranking of the starter is avoided by introducing calculated delays in between successive cranking.

4.3 Fuel savers:

Stop-start systems can cut combined city-highway fuel consumption and greenhouse gas emissions by 3-10 percent, and even more when they're combined with hybrid drive systems. By eliminating engine idling, stop-start systems also reduce toxic and smog-causing tailpipe emissions. The systems deliver their fuel economy improvements in city situations that involve lots of stop-and-go driving. And they can do it at relatively little cost to the consumer. While stop-start systems have typically been included in the overall price premium for hybrids, many industry analysts say their actual cost is between \$300 and \$400.

4.4 Consumer gain:

With gasoline at \$4 a gallon, a \$300 stop-start system could pay for itself in three years or less — a payback period most U.S. consumers find quite acceptable. A 20-mpg car outfitted with a system that offers a 3.5 percent fuel economy improvement over 15,000 miles of annual driving would use almost 25.5 fewer gallons of gas each year. That's an annual savings of \$102. A \$300

system offering a 10 percent fuel economy boost would earn back its cost in just 13 months with gas at \$4 a gallon.

4.5 Recent trends:

Although the savings you'll get on fuel in a mild hybrid won't really compare to those of a full hybrid car, stop-start technology is still a promising step forward in making conventional cars much more fuel-efficient. This is an especially important step, since most fuel economy problems stem from idling and the constant stop and go nature of city driving. The technology has actually been around for a while, but we'll take a look at some notable newer cars using idle-stop systems and the improvements that are underway for the existing technology.

Several recent car models will use (or have used) some kind of stop-start technology, with most examples coming from Europe or Japan. Some people might not know that the MINI Cooper, built under the BMW Group, has actually used idle-stop technology since 2007. And German automaker Audi started using an idle-stop system in the second quarter of 2009 in its A3, A4 and A5 models, with more to follow.

The most recent, and perhaps the most significant, example of a vehicle receiving an idle-stop system is the Mazda3, which will begin using a new and improved type of stop-start technology by the end of 2009. While some systems tend to suffer due to the use of a conventional electric starter, which can cause slow and sluggish restarts, the new Mazda system will use combustion for the restart. Direct injection plays an integral part in the function of the system that Mazda calls the Smart Idle Stop System, or SISS. When a vehicle slows down and stops, sensors position the engine's pistons in specific locations within each of the cylinders. This allows the

system to determine which of the cylinders is fully pressurized and ready to fire. At restart, fuel is injected into the appropriate cylinder, the fuel and air mix is ignited and the engine continues to operate as it normally would.

With these slight improvements, Mazda claims that their engines can restart in 350 milliseconds, about half the time it takes for other conventional systems to restart, and it results in a 10 percent reduction in fuel use, too.

Ford recently announced that its stop-start system, available on some 2012 cars and SUVs, has the real-world potential to boost fuel economy by “as much as 10 percent.” Mazda, more concerned with EPA results, says a 3 with i-stop gains only one-tenth of one percent on the EPA city cycle. Therefore, U.S. Mazdas will do without stop-start until it can be part of a worthwhile package of efficiency improvements. (The Japanese city-cycle test has 40 seconds of idling, so the cost of i-stop is justified there.)

4.6 Stop-Start Downsides:

There's a small learning curve involved in driving a vehicle with a stop-start system. Many drivers find it disconcerting when the engine shuts off each time they come to a stop. And some systems aren't as smooth as others, with drivers reporting unexpected jerks and jolts when the engine restarts. BMW's system, for example, has been widely criticized for this. Many owners report that they have disabled the system with the shutoff switch that's standard with the BMW system. Indeed, most stand-alone stop-start systems come with driver-selectable on-off switches.

"Education needs to accompany introduction of this technology because car buyers will have to learn how these systems function differently from what they are used to," says Edmunds.com

Senior Analyst Jessica Caldwell. "The burden for this will fall on the dealerships, but smart dealers will see it as another way to build relationships with consumers." Finally, if the special 12-volt batteries needed for stand-alone stop-start systems do need replacement, they are about twice as expensive as conventional 12-volt lead-acid batteries.

CHAPTER 5

Ultra-capacitor

5.1 Introduction:

Ultra capacitors are a new technology that allows storing 20 times more energy than conventional electrolytic capacitors. Despite this important advance in energy storage, they are still far from being compared with electrochemical batteries. Even Lead-acid batteries can store at least ten times more energy than ultra capacitors. However, they present a lot better performance in specific power than any battery, and can be charged and discharged thousands of times without performance deterioration. These very good characteristics can be used in combination with normal electrochemical batteries, to improve the transient performance of an electric vehicle, and to increase the useful life of the batteries. Fast and sudden battery discharge during acceleration, or fast charge during regenerative braking can be avoided with the help of ultra capacitors. Besides, ultra capacitors allow regenerative braking even with the batteries fully charged.

5.2 Capacitor:

A basic capacitor consists of two metal plates, or conductors (typically aluminum), separated by an insulator, such as air or a film made of plastic, or ceramic. During charging, electrons accumulate on one conductor, and depart from the other. In effect, a negative charge builds on one side while a positive charge builds on the other.

The negatively charged electrons *want* to join the depleted (positive) side, but can't cross over that non-conductive insulator (for the most part, anyway—there is some leakage). This separation of positive and negative charges, which want to balance out, or neutralize, each other, creates what's called an electric field. Discharging occurs when the electrons are given a path to flow to the other side—in other words, when balance is restored.

5.3 Putting the “ultra” in ultra capacitors:

Ultra capacitors also have two metal plates, but they are coated with a sponge-like, porous material known as activated carbon. And they're immersed in an electrolyte made of positive and negative ions dissolved in a solvent. One carbon-coated plate, or electrode, is positive, and the other is negative. During charging, ions from the electrolyte accumulate on the surface of each carbon-coated plate.

Like capacitors, ultra capacitors store energy in an electric field, which is created between two oppositely charged particles when they are separated. In an ultracapacitor, we have this electrolyte, in which an equal number of positive and negative ions are uniformly dispersed. And remember that in a capacitor, negative charge builds on one side and positive charge builds on the other. Similarly, in an ultracapacitor, when voltage is applied across the two metal plates (i.e. during charging), a charge still builds on the two electrodes—one positive, one negative. This then causes each electrode to attract ions of the opposite charge. For an ultracapacitor, each carbon electrode ends up having *two layers* of charge coating its surface (thus, ultracaps are also called “double layer capacitors”)

5.4 The ultra cap advantage:

Unlike capacitors and ultra capacitors, batteries store energy in a chemical reaction. Ions are actually inserted into the atomic structure of an electrode (in an ultra cap, the ions simply cling). This is an important distinction, because storing energy *without chemical reactions* allows ultra capacitors to charge and discharge much faster than batteries. And because capacitors don't suffer the wear and tear caused by chemical reactions, they can also last much longer. Charge separation is at work in both capacitors and ultra capacitors. But in a capacitor, the separated charges can get no closer than the distance between the two metal plates. They're awfully close together—on the order of tens of microns—but limited by the thickness of that ceramic or paper film in the middle (one micron is one-thousandth of a millimeter). In an ultracapacitor, the distance between the ions and opposite-charged electrode is so tiny it's measured in nanometers (one-thousandth of a micron). The shorter distance between those separated charges in an ultracapacitor translates to a larger electric field—and much more energy storage capacity.

That's only part of why ultra capacitors can store more energy than regular capacitors. The activated carbon is also a key. It affords a surface area 10,000 to 100,000 times greater than the linear surface area of the naked metal. Put simply, all those nooks and crannies in the surface allow more ions to cling to the electrode.

5.5 Measuring capacitance:

Surface area makes a huge difference for what's called capacitance, or the amount of electric charge a device will hold given a certain amount of voltage. Capacitance is the key metric for

comparing capacitor performance, and it's measured in Farads (named, as *Lost* fans might appreciate, after the chemist and physicist Michael Faraday).

Now, the Farad is such a huge unit of measurement, "it's like measuring distance in light years". So it's much more common to see microfarads (one-*millionth* of a farad) and even pico farads (one-millionth of a microfarad).

A capacitor the size of a D-cell battery, for example, has a capacitance of only about 20 microfarads. But a similarly sized ultracapacitor has a capacitance of 300 Farads. That means, at the same voltage, the ultracapacitor could in theory store up to 15 million times more energy than the capacitor.

Here is where we run into some of the challenges with ultra capacitors, however. A typical 20-microfarad capacitor would be able to handle as much as 300 volts, while an ultra cap would be rated at only 2.7 volts. At a higher voltage, the electrolyte starts to break down. So realistically we're talking about an ultracapacitor storing about 1,500 times the energy of a comparably sized capacitor.

5.6 Ultra caps and batteries as partners:

Despite offering a huge leap over regular capacitors, ultra caps still lag behind batteries when it comes to energy storage capacity. Ultra capacitors (which are also more expensive per energy unit than batteries), can store only about 5 percent of the energy of comparable lithium-ion batteries.

Ultra capacitors are very effective, however, at accepting or delivering a sudden surge of energy, and that makes them a good partner for lithium-ion batteries. In an electric car, for example, an ultracapacitor could provide the power needed for acceleration, while a battery provides range and recharges the ultra cap between surges.

The ultracapacitor is like a small bucket with a big spout. Water can flow in or out very fast, but there's not very much of it. The battery is like a big bucket with a tiny spout. It can hold much more water, but it takes a long time to fill and drain it. The small bucket can provide a brief "power surge" ("lots of water" in this analogy), and then refill gradually from the big bucket.

5.7 Putting ultra caps to work:

The devices appear in hundreds of applications, from cell phone base stations to alarm clocks (as backup power) to audio systems.

Ultra capacitors hold promise for a similar job on the electric grid. Today, transmission lines operate below full capacity (often somewhere above 90 percent in order to leave a buffer for power surges. Banks of ultra capacitors could be set up to absorb power surges, enabling transmission lines to run closer to 100 percent capacity.

It might not seem like much, especially considering that it would take warehouse-sized banks for ultra caps to do the job. But installing ultra capacitors to handle the peaks would actually be much cheaper than adding even 5 percent more capacity with new transmission lines.

In cars, ultra capacitors could play a role in the growing market for "micro hybrids," which cut the engine during idling. In these "start-stop" systems , "The ultracapacitor would provide

power during the stop (lights, radio, air conditioner, etc.).” It would also provide power for the restart, and then be “recharged during the next interval of travel.”

5.8 Recent trends:

According to a new report from Innovative Research and Products (iRAP) titled “Ultra capacitors – A Global Industry and Market Analysis (ETP-101),” the ultra capacitors market will see continued growth during 2006 to 2011. The \$272 million (in US\$) worldwide ultracapacitor business in 2006 will continue to grow at an AAGR of 15.3% through 2011 to reach \$560 million.

The ultra capacitor market has grown through application to consumer electronics. The most typical case is mobile phone applications such as memories and time clocks. Based on these applications, the global ultra capacitor market recorded 480 billion won in 2011. However, it is expected that the market growth will be led by transportation applications in the future, reaching 840 billion won in 2015 and 1 trillion 250 billion won in 2012 at the annual growth rate of 11.3%.

CHAPTER 6

Regenerative Braking:

6.1 Introduction

Every time we step on the car's brakes, we are wasting energy. Physics tells us that energy cannot be destroyed. So when our car slows down, the kinetic energy that was propelling it forward has to go somewhere. Most of it simply dissipates as heat and becomes useless. That energy, which could have been used to do work, is essentially wasted.

In most cars it's the inevitable byproduct of braking and there's no way we can drive a car without occasionally hitting the brakes. A lot of thought have been given to this aspect and a braking system that can recapture much of the car's kinetic energy and convert it into electricity, so that it can be used to recharge the car's batteries or can be used for some other purpose has been developed. This system is called regenerative braking.

6.2 Definition

Regenerative braking is a system in which the electric motor that normally drives a hybrid or pure electric vehicle is essentially operated in reverse (electrically) during braking or coasting. Instead of consuming energy to propel a vehicle, the motor acts as a generator that charges the onboard batteries with electrical energy that would normally be lost as heat through traditional mechanical friction brakes. As the motor “acts in reverse,” it generates electricity. The

accompanying friction (electrical resistance) assists the normal brake pads in overcoming inertia and helps slow the vehicle.

6.3 Traditional braking vs. Regenerative braking

In a traditional braking system, brake pads produce friction with the brake rotors to slow or stop the vehicle. Additional friction is produced between the slowed wheels and the surface of the road. This friction is what turns the car's kinetic energy into heat. With regenerative brakes, on the other hand, the system that drives the vehicle does the majority of the braking. When the driver steps on the brake pedal of an electric or hybrid vehicle, these types of brakes put the vehicle's electric motor into reverse mode, causing it to run backwards, thus slowing the car's wheels. While running backwards, the motor also acts as an electric generator, producing electricity that's then fed into the vehicle's batteries. These types of brakes work better at certain speeds than at others. In fact, they're most effective in stop-and-go driving situations. However, hybrids and fully electric cars also have friction brakes, as a kind of back-up system in situations where regenerative braking simply won't supply enough stopping power. In these instances, it's important for drivers to be aware of the fact that the brake pedal might respond differently to pressure. The pedal will sometimes depress farther towards the floor than it normally does and this sensation can cause momentary panic in drivers.

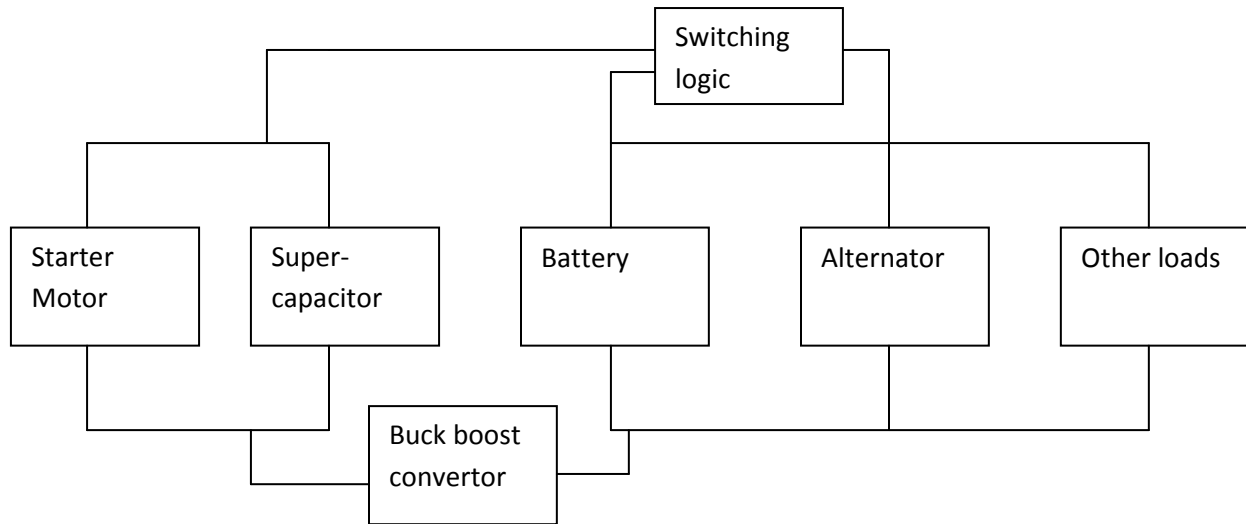
6.4 Regenerative braking circuits

Regenerative braking is used in vehicles that make use of electric motors, primarily fully electric vehicles and hybrid electric vehicles. One of the more interesting properties of an electric motor is that, when it's run in one direction, it converts electrical energy into mechanical energy that

can be used to perform work (such as turning the wheels of a car), but when the motor is run in the opposite direction, a properly designed motor becomes an electric generator, converting mechanical energy into electrical energy. This electrical energy can then be fed into a charging system for the car's batteries.

In a regenerative braking system, the trick to getting the motor to run backwards is to use the vehicle's momentum as the mechanical energy that puts the motor into reverse. Momentum is the property that keeps the vehicle moving forward once it's been brought up to speed. Once the motor has been reversed, the electricity generated by the motor is fed back into the batteries, where it can be used to accelerate the car again after it stops. Sophisticated electronic circuitry is necessary to decide when the motor should reverse, while specialized electric circuits route the electricity generated by the motor into the vehicle's batteries. In some cases, the energy produced by these types of brakes is stored in a series of capacitors for later use. In addition, since vehicles using these kinds of brakes also have a standard friction braking system, the vehicle's electronics must decide which braking system is appropriate at which time. Because so much is controlled electronically in a regenerative braking system, it's even possible for the driver to select certain presets that determine how the vehicle reacts in different situations. For instance, in some vehicles a driver can select whether regenerative braking should begin immediately whenever the driver's foot comes off the accelerator pedal and whether the braking system will take the car all the way to 0 mph (0 kilometers per hour) or will let the car coast slightly.

6.5 Regenerative Braking Controller:



Brake controllers are electronic devices that can control brakes remotely, deciding when braking begins ends, and how quickly the brakes need to be applied. In towing situations, for instance, brake controllers can provide a means of coordinating the brakes on a trailer with the brakes on the vehicle doing the towing.

The battery is assumed to be fully charged in this case. It is normally connected in parallel with the alternator and the load. During regeneration, battery will no longer be connected to alternator. The battery connection is given to the buck-boost convertor circuit in order to maintain the energy flow with the super-capacitor during the capacitor's charging and discharging times.

During the initial start of the engine, the super-capacitor charge status is checked. This has to well above the minimum value required to start the motor.

During normal vehicle running condition, the super-capacitor has to be kept in between the minimum and the maximum threshold values. This is an essential step in order to meet the two

ends. One is that, it must have sufficient energy to start the motor and the other is that it must be kept minimum in order to absorb considerable amount of energy during braking. This is achieved by the use of buck-boost convertors in between the battery and super-capacitor.

During braking condition, if the vehicle has reached a certain threshold speed and if clutch is not pressed, super-capacitor is been supplied from the alternator. This is carried out by the switching circuit logic in the network. This charging of the capacitor continues till the capacitor can reach a maximum threshold limit. If this is exceeded, and if the brake event has not stopped, then the buck convertor is activated. This enables the battery also to take part in regeneration process.

After the braking event has occurred, the super-capacitor has to be discharged immediately in order to allow further energy recovery for successful braking. This is done through the buck convertor in the circuit.

The inclusion of auto-stop feature introduces an added advantage to the system. In this case, the super-capacitor has to be charged at the instant when the engine is turned off due to auto stop feature.

The most important function of the brake controller, however, may be deciding whether the motor is currently capable of handling the force necessary for stopping the car. If it isn't, the brake controller turns the job over to the friction brakes, averting possible catastrophe. In vehicles that use these types of brakes, as much as any other piece of electronics on board a hybrid or electric car, the brake controller makes the entire regenerative braking process possible.

6.6 Regenerative Braking Efficiency

The energy efficiency of a conventional car is only about 20 percent, with the remaining 80 percent of its energy being converted to heat through friction. The miraculous thing about regenerative braking is that it may be able to capture as much as half of that wasted energy and put it back to work. This could reduce fuel consumption by 10 to 25 percent. Hydraulic regenerative braking systems could provide even more impressive gains, potentially reducing fuel use by 25 to 45 %. In a century that may see the end of the vast fossil fuel reserves that have provided us with energy for automotive and other technologies for many years, and in which fears about carbon emissions are coming to a peak, this added efficiency is becoming increasingly important.

The beginning of the 21st century could very well mark the final period in which internal combustion engines are commonly used in cars. Already automakers are moving toward alternative energy carriers, such as electric batteries, hydrogen fuel and even compressed air. Regenerative braking is a small, yet very important, step toward our eventual independence from fossil fuels. These kinds of brakes allow batteries to be used for longer periods of time without the need to be plugged into an external charger. These types of brakes also extend the driving range of fully electric vehicles. Sure, these cars may use fossil fuels at the recharging stage -- that is, if the source of the electricity comes from a fossil fuel such as coal -- but when they're out there on the road, they can operate with no use of fossil fuels at all, and that's a big step forward. The added efficiency of regenerative braking also means less pain at the pump, since hybrids with electric motors and regenerative brakes can travel considerably farther on a gallon of gas, some achieving more than 50 miles per gallon at this point.

CHAPTER 7

Controller

7.1 Introduction

The controller used in this project is the TMS320F28035. The F28035 Piccolo family of microcontrollers provides the power of the C28035 core and Control Law Accelerator (CLA) coupled with highly integrated control peripherals in low pin-count devices. This family is code-compatible with previous C28035-based code, as well as providing a high level of analog integration. An internal voltage regulator allows for single rail operation. Enhancements have been made to the HRPWM module to allow for dual-edge control (frequency modulation). Analog comparators with internal 10-bit references have been added and can be routed directly to control the PWM outputs. The ADC converts from 0 to 3.3-V fixed full scale range and supports ratio-metric VREFHI/VREFLO references. The ADC interface has been optimized for low overhead/latency.

7.2 Key features

Highlights

- High-Efficiency 32-Bit CPU
- 60-MHz Device Reset
- Single 3.3-V Supply
- Integrated Power-on and Brown-out Resets

- Two Internal Zero-pin Oscillators
- Up to 45 Multiplexed GPIO Pins
- Three 32-Bit CPU Timers
- On-Chip Flash, SARAM, OTP Memory
- Code-Security Module
- Serial Port Peripherals (SCI/SPI/I2C/LIN/eCAN)
- Enhanced Control Peripherals
 - Enhanced Pulse Width Modulator (ePWM)
 - High-Resolution PWM (HRPWM)
 - Enhanced Capture (eCAP)
 - Enhanced Quadrature Encoder Pulse
 - Analog-to-Digital Converter (ADC)
 - On-Chip Temperature Sensor
 - Comparator

High-Efficiency 32-Bit CPU

- 60 MHz (16.67-ns Cycle Time)
- 16 x 16 and 32 x 32 MAC Operations
- 16 x 16 Dual MAC
- Harvard Bus Architecture
- Atomic Operations
- Fast Interrupt Response and Processing
- Code-Efficient (in C/C++ and Assembly)
- Programmable Control Law Accelerator (CLA)

- 32-Bit Floating-Point Math Accelerator
- Executes Code Independently of the Main CPU
- Low Device and System Cost
- Single 3.3-V Supply
- Integrated Power-on Reset and Brown-out Reset
- Low Power
- No Analog Support Pins

Clocking:

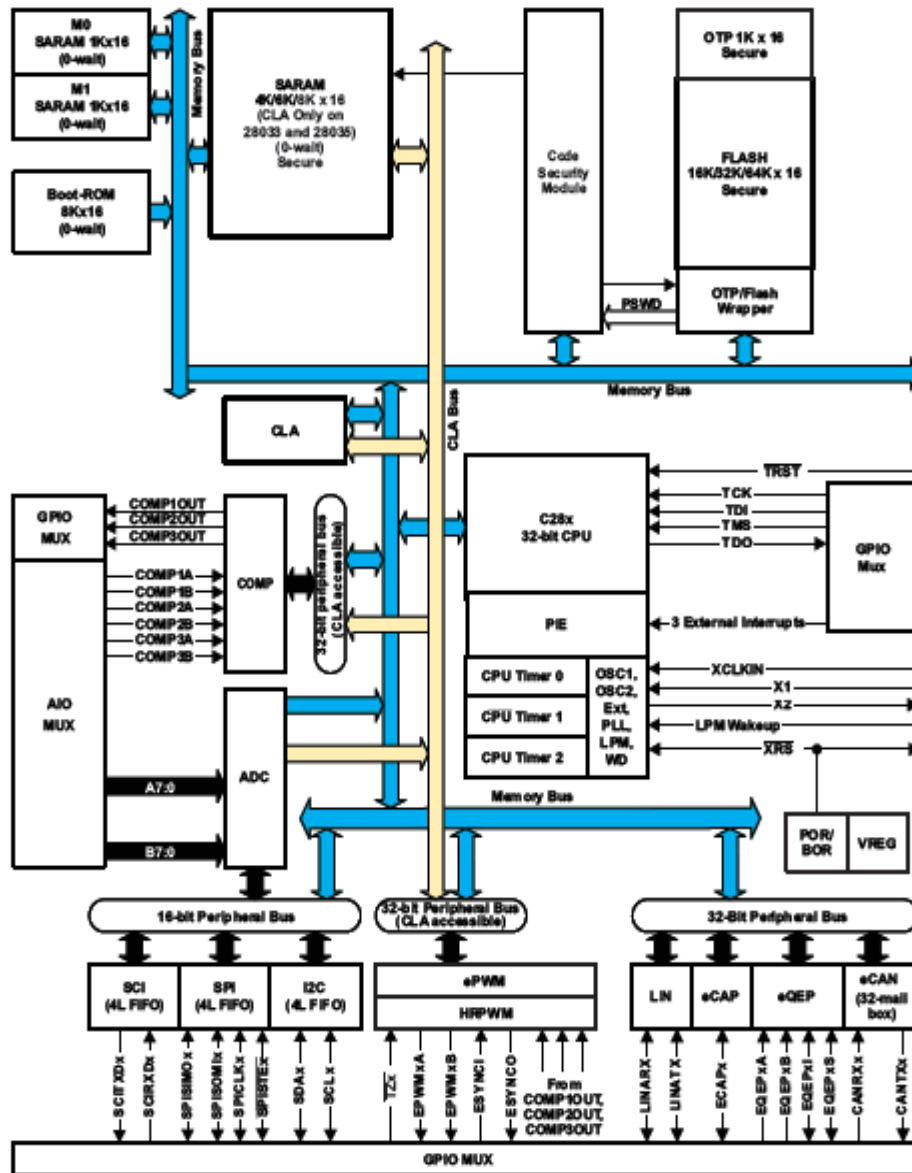
- Two Internal Zero-pin Oscillators
- On-Chip Crystal Oscillator/External Clock input
- Dynamic PLL Ratio Changes Supported
- Watchdog Timer Module
- Up to 45 Individually Programmable, Multiplexed GPIO Pins With Input Filtering

Serial Port Peripherals

- One SCI (UART) Module
- Two SPI Modules
- One Inter-Integrated-Circuit (I2C) Bus
- One Local Interconnect Network (LIN) Bus
- One Enhanced Controller Area Network

Advanced Emulation Features

7.3 BLOCK DIAGRAM:



7.4 GPIO MUX

The GPIO MUX can multiplex up to three independent peripheral signals on a single GPIO pin in addition to providing individual pin bit-banging I/O capability.

7.5 JTAG PORT

On the 2803x device, the JTAG port is reduced to 5 pins (TRST, TCK, TDI, TMS, and TDO). TCK, TDI, TMS and TDO pins are also GPIO pins. The TRST signal selects either JTAG or GPIO operating mode for the pins. During emulation/debug, the GPIO function of these pins is not available. If the GPIO38/TCK/XCLKIN pin is used to provide an external clock, an alternate clock source should be used to clock the device during emulation/debug since this pin will be needed for the TCK function.

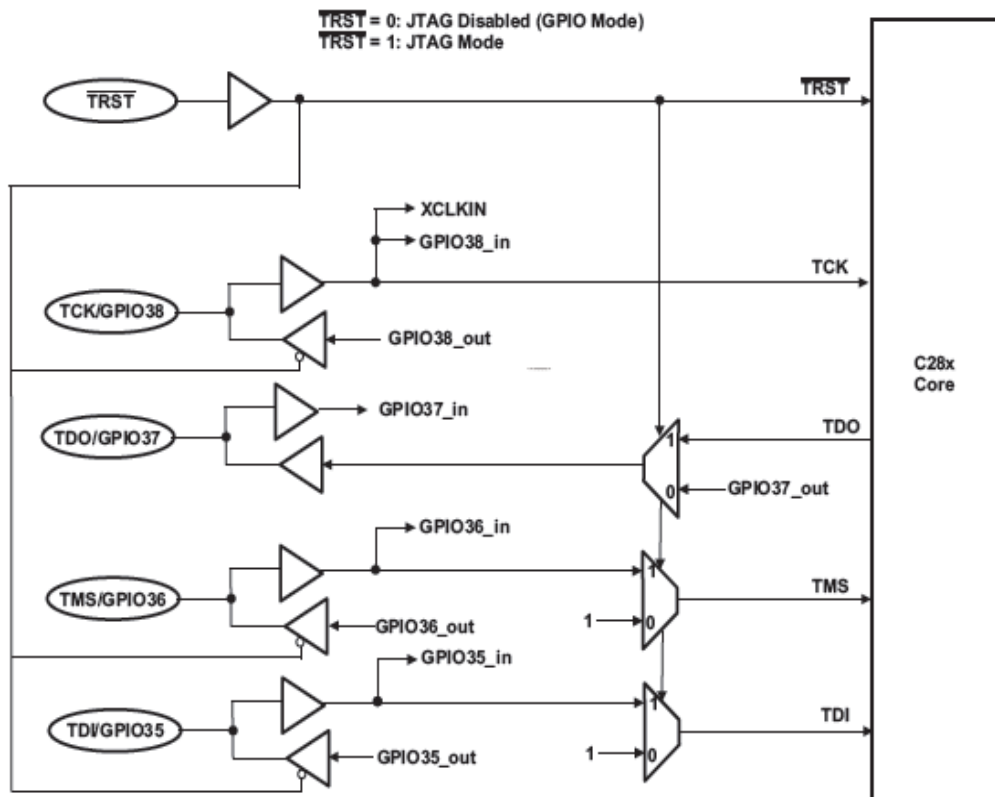


Figure 4-16. JTAG/GPIO Multiplexing

7.6 ADC

The core of the ADC contains a single 12-bit converter fed by two sample-and-hold circuits. The sample-and-hold circuits can be sampled simultaneously or sequentially. These, in turn, are fed by a total of up to 16 analog input channels. The converter can be configured to run with an internal band gap reference to create true-voltage based conversions or with a pair of external voltage references (VREFHI/VREFLO) to create ratio metric-based conversions.

Contrary to previous ADC types, this ADC is not sequencer-based. It is easy for the user to create a series of conversions from a single trigger. However, the basic principle of operation is centered on the configurations of individual conversions, called SOC's, or Start-Of-Conversions.

Functions of the ADC module include:

- 12-bit ADC core with built-in dual sample-and-hold (S/H)
- Simultaneous sampling or sequential sampling modes
- Full range analog input: 0 V to 3.3 V fixed, or VREFHI/VREFLO ratio metric. The digital value of the input analog voltage is derived by:

Internal Reference (VREFLO = VSSA. VREFHI must not exceed VDDA when using either internal or external reference modes.)

Digital value = 0 When input ≤ 0 V

Digital value = $(4096 * (\text{Input analog Voltage} - \text{VREFLO})) / 3.3$. When $0 < \text{input} < 3.3 \text{ V}$

Digital value = 4095 When input ≥ 3.3 V

External Reference (VREFHI/VREFLO connected to external references. VREFHI must not exceed VDDA when using either internal or external reference modes.)

Digital value = 0 When input ≤ 0 V

Digital value =

$(4096 * (\text{Input analog Voltage} - \text{VREFLO})) / (\text{VREFHI} - \text{VREFLO})$ When $0 < \text{input} < \text{VREFHI}$

Digital value = 4095 When input ≥ VREFHI

- Runs at full system clock, no pre scaling required
- Up to 16-channel, multiplexed inputs
- 16 SOC's, configurable for trigger, sample window, and channel
- 16 result registers (individually addressable) to store conversion values
- Multiple trigger sources
 - S/W – software immediate start
 - EPWM 1–7
 - GPIO XINT2
 - CPU Timers 0/1/2
 - ADCINT1/2
- 9 flexible PIE interrupts, can configure interrupt request after any conversion.

CHAPTER 8

Hardware architecture

8.1 Introduction

The hardware tools used for this project are

- USB docking station
- TMS320F28035 control card
- Start- stop control PCB
- General purpose board for Regen
- Power section with switches
- Relays

Other sources and sinks for testing at laboratory level

- Variable Dual power supply
- Function generators
- Oscilloscope

8.2 Description:

USB docking station:

This is an Experimenter Kit from Texas Instruments. The docking station has access to all control CARD signals, breadboard areas, RS-232, JTAG connector, and on board USB JTAG

emulation. No JTAG emulator required, as docking station features on board USB JTAG emulation. It also contains 5V and 3.3V supply. This can be powered either through a 5 V supply or through USB. In this project, it is used to interface the code composer studio and the controller through the on-board USB JTAG.

Control card:

Control CARDS for TMS320C2000 put the microcontroller and all necessary support devices in a standard, pin-compatible DIM-socket daughter card. With a common plug-in compatible interface, control CARDS allow users to quickly experiment with various TMS320C2000 MCUs application by simply plugging in a new control CARD.

Start-stop PCB:

This board contains all the signal conditioning circuits required for the system. Potential dividers for digital inputs, buffer circuits for analog inputs and hysteresis network for the pulsed inputs are used. For output signal conditioning, relay drivers are used. Both low side and high side drivers are used for switching purpose.

GP board for Regen:

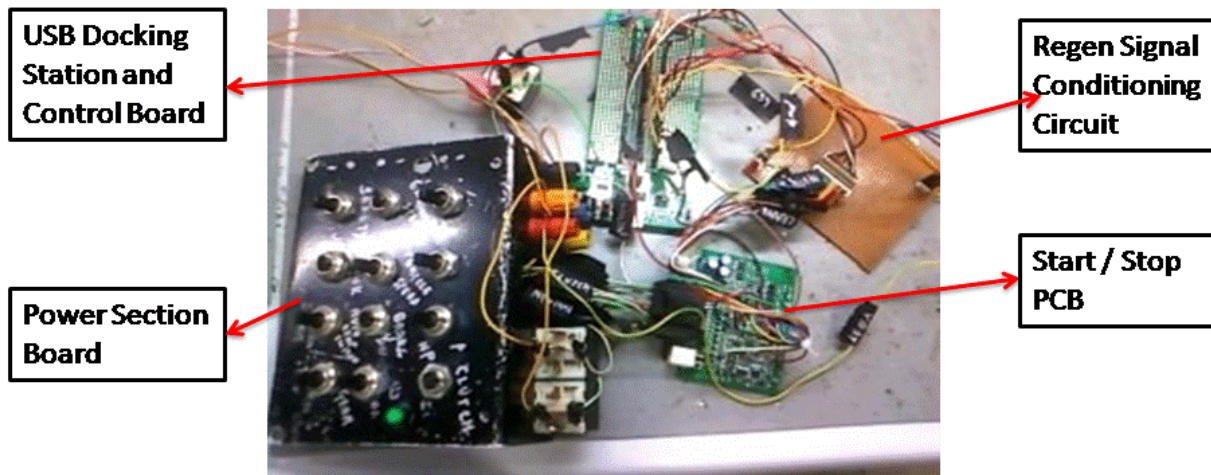
The general purpose board contains the output signal conditioning for the three Regenerative braking signals. It consists of transistors to drive the corresponding relays. It also consists of the Regenerative braking LED. i.e., the LED used to indicate the charging status of super-capacitor.

Power section and relays:

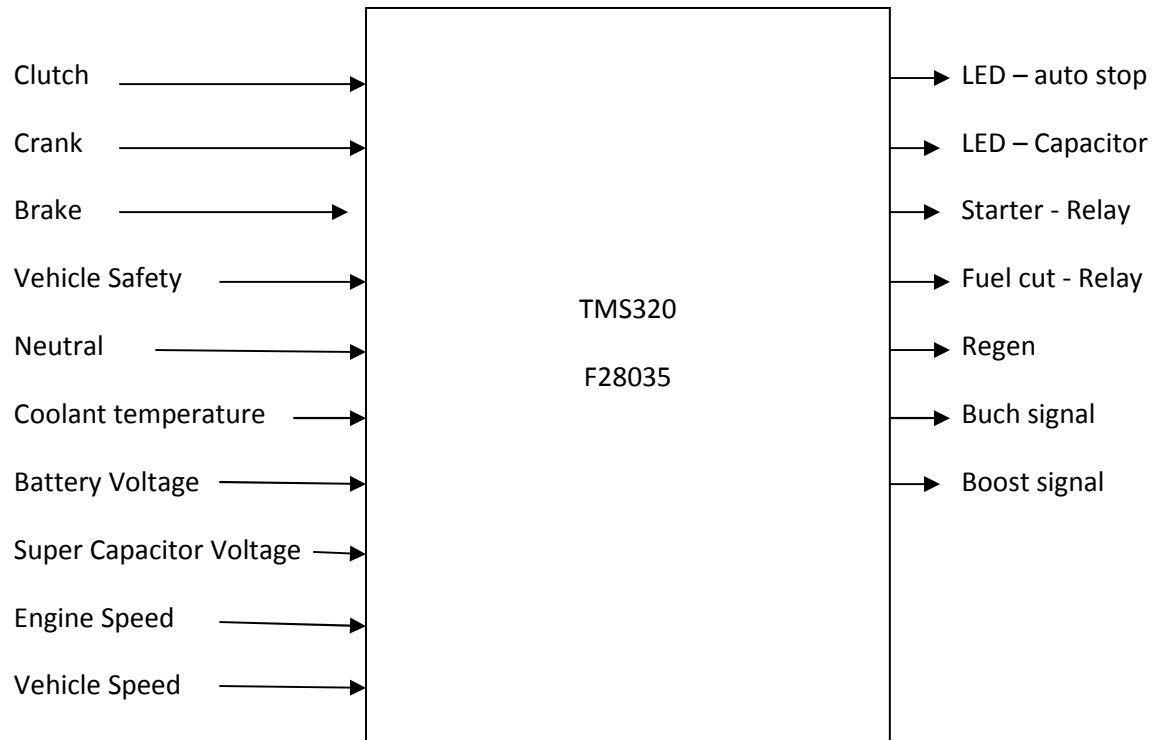
This consists of the various switches and power supply and relays. These are used to give input to the hardware circuitry.

The dual supply is used to provide simulation signals corresponding to battery voltage, super-capacitor voltage and coolant temperature. The function generator provides engine and vehicle speed pulses. The oscilloscope is used to monitor the output signals.

The following gives the overall picture of the hardware used at laboratory level.

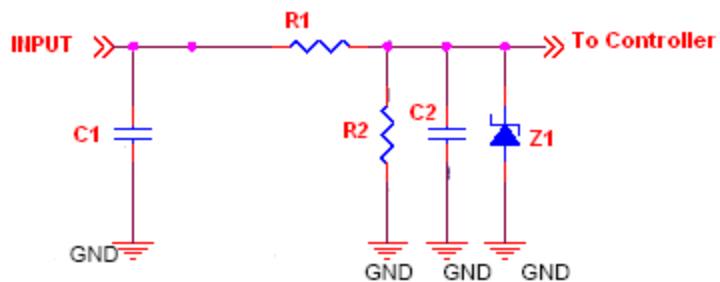


8.3 Block diagram



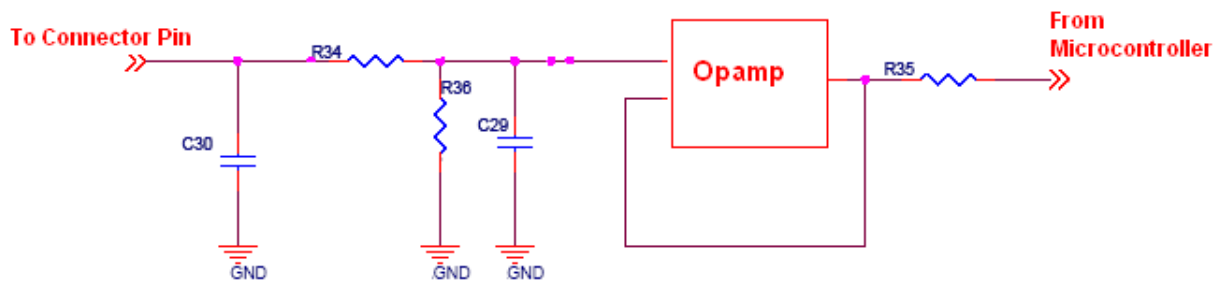
8.4 Signal conditioning

Digital inputs



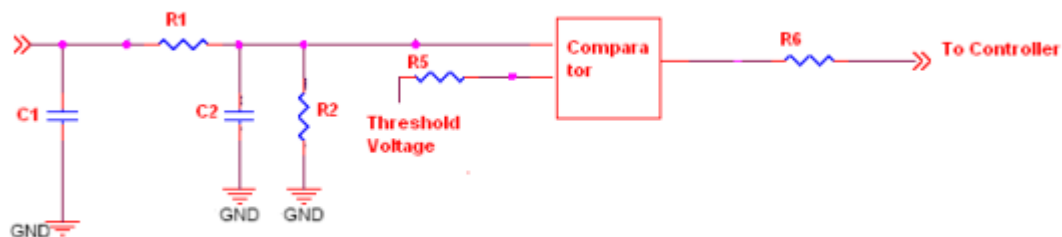
The resistors are selected in such a way to provide a drop of 3.3 volt for the maximum input range and also to provide a drop of 1.5 volts as logic high for the minimum input range. A 3.3 volt zener is been used as a protective measure before giving to the controller. The capacitors are provided to arrest the noise and transient voltages. The digital inputs used in this project are clutch, crank, vehicle safety, neutral sensor, brake and auto stop enable disable switch.

Analog inputs



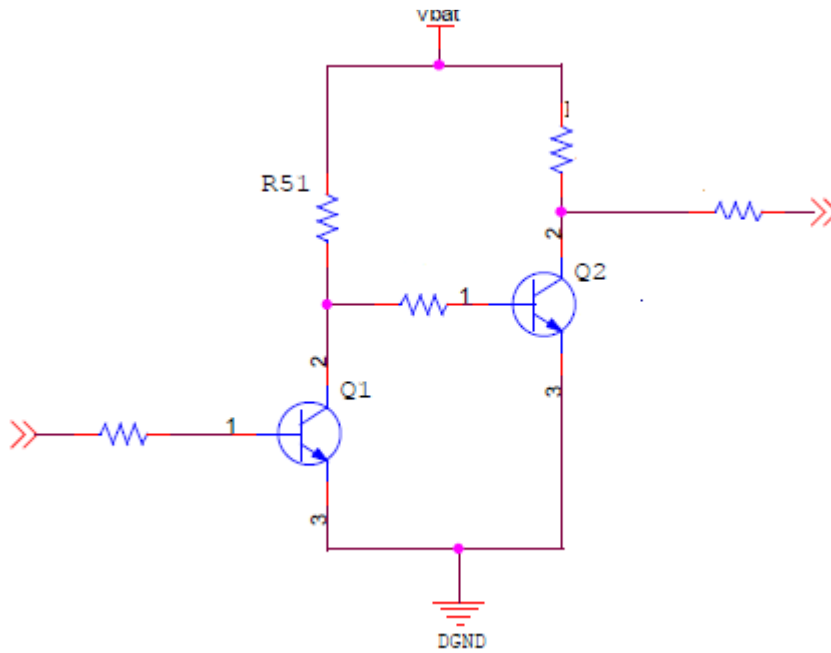
The op-amp acts as a buffer to give a steady value of the input voltage. The resistor divider section is used to provide a maximum of 3.3 volt drop. The analog inputs used in the circuit are coolant temperature, super capacitor voltage, battery voltage.

Pulsed inputs



This network forms a hysteresis loop. This ensures the suppression of noise in the circuit and gives the pulsed inputs whose frequency corresponds to pre-determined speed. Pulsed inputs used in this circuit are engine speed and vehicle speed.

Relay drivers



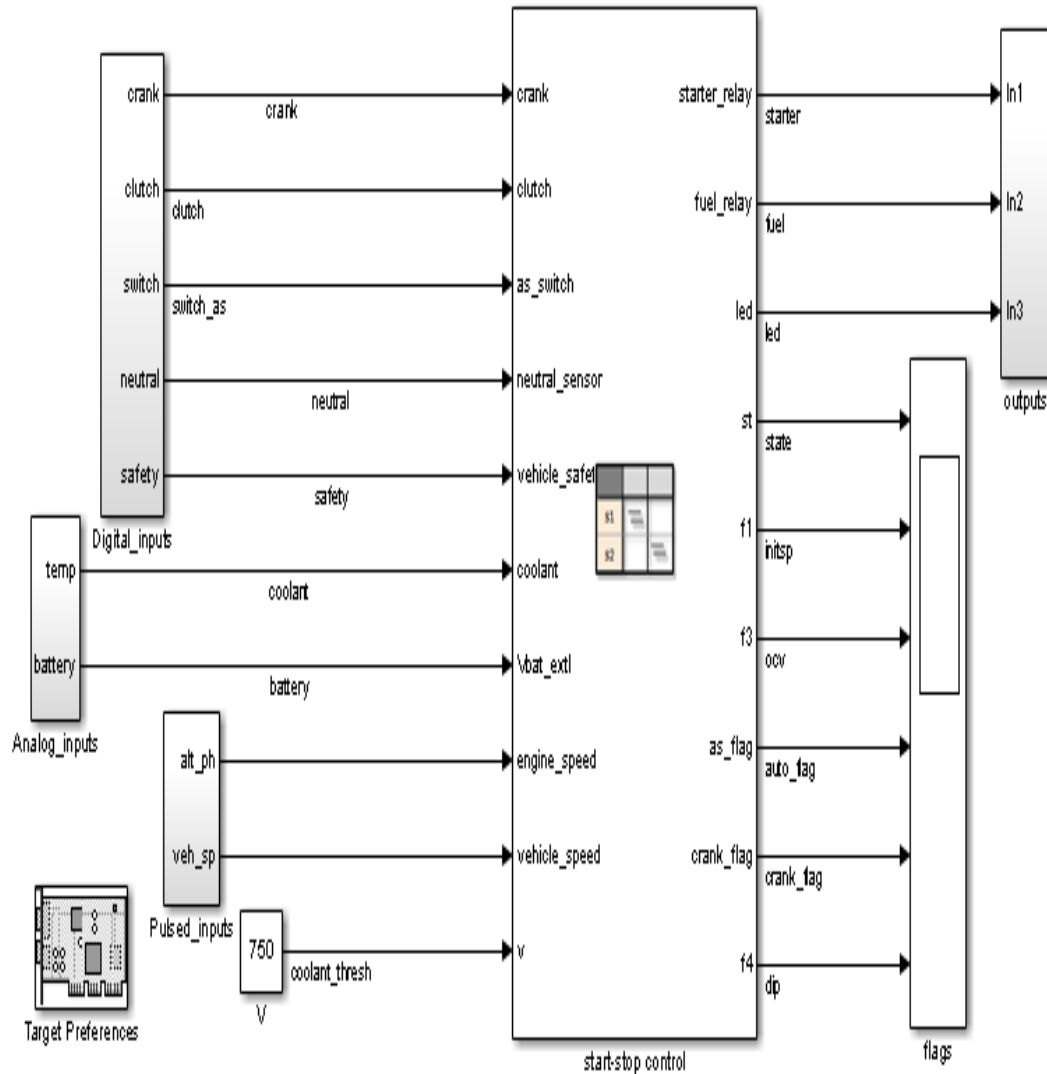
The low voltage signals from controller are stepped up using the above driver section.

CHAPTER 9

Software architecture

9.1 Stop-start control

The software tools used are the Simulink – state flow and the Code composer Studio.

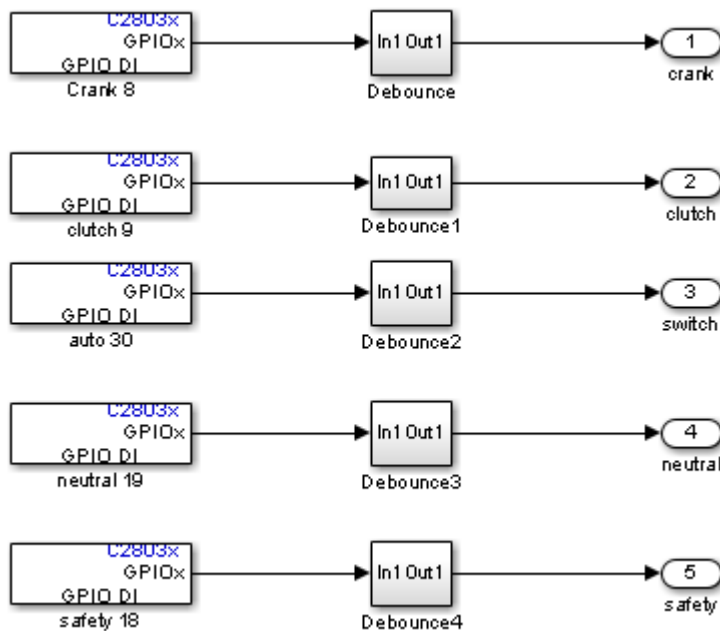


Major blocks:

1. Digital inputs
2. Analog inputs
3. Pulsed inputs
4. Target Preferences
5. Start-stop control
6. Outputs
7. Flags

9.1.1 DIGITAL INPUTS:

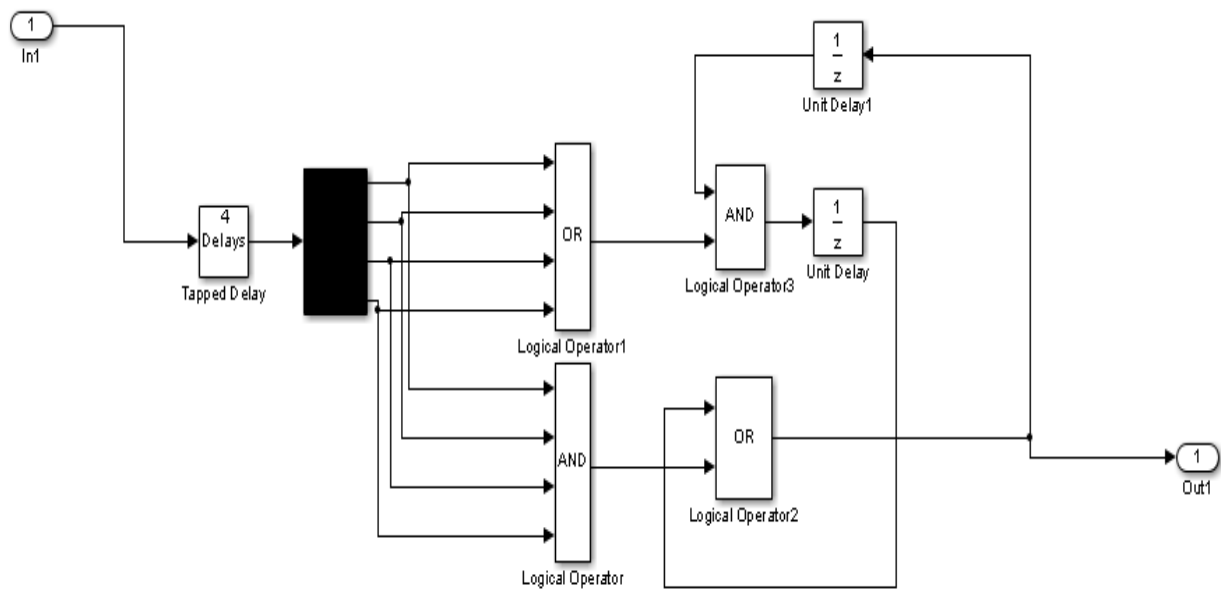
This block contains the digital inputs required for the system. When expanded,



It has the five digital input blocks. Each block has been set as general purpose input pin. The logic indication is as follows.

SIGNAL	LOGIC HIGH	LOGIC LOW
Crank	The user turns on the car key to start the engine.	The user turns off the car key.
Clutch	Pressed	Released
Auto-endis	Enabled	Disabled
Neutral sensor	Gear in neutral	Gear not in neutral
Vehicle safety	Bonnet is closed	Bonnet is open

Each of the five inputs has been connected to a block called-”Debounce”. The significance or the necessity of this debounce block is that, when we use switches, it is bound to suffer from glitches. As a result of this, it does not reach a steady ON or OFF state. In order to avoid these glitches, software debounce has been achieved. This logic samples 4 values in a buffer manner and check for steady state. If steady state is not reached, the previous value is retained else, the new value is accepted. The expansion of the block is as follows:



The above block samples 4 values of the digital input periodically and performs the debounce formula to give the steady output.

Debounce formula:

Output & =m [1] | m [2] | m [3] | m [4];

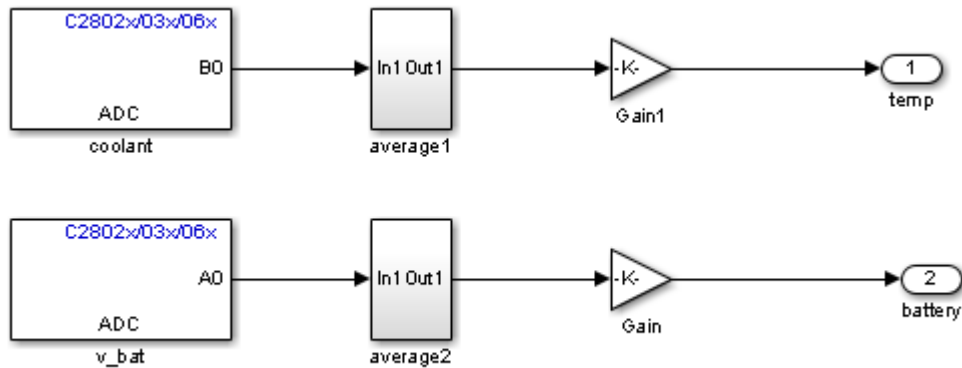
Output | =m [1] & m [2] & m [3] & m [4];

Here, m [1], m [2], m [3] and m [4] are values of the input sampled continuously.

The same block is implemented for all the digital inputs.

9.1.2 ANALOG INPUTS:

This block contains the analog inputs required for the system. When expanded,

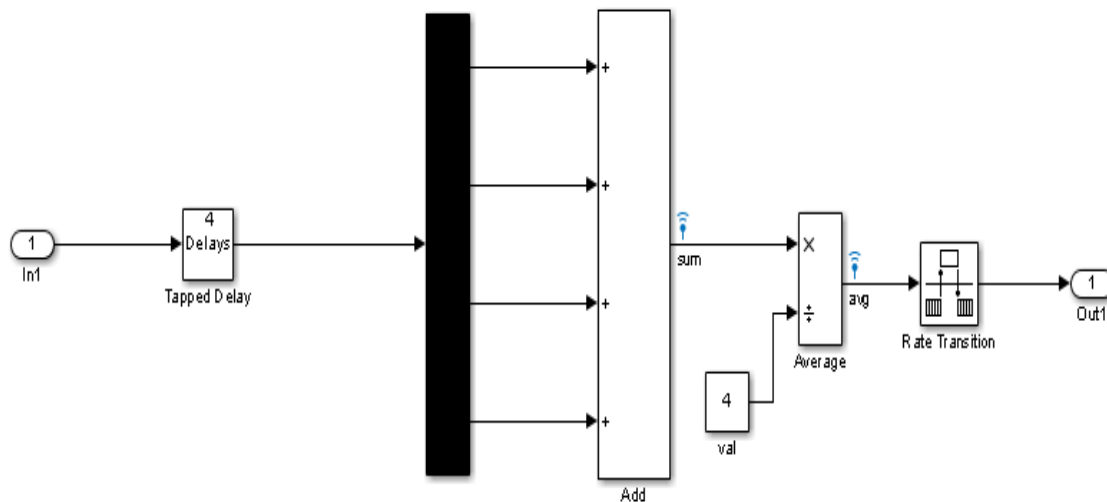


It has two analog inputs given to the ADC block of the controller. The ADC block's configuration is set to work in simultaneous mode. The input range for the two inputs is as follows:

SIGNAL	RANGE
Coolant temperature	0 to 4.5 volts. Signal conditioned as from 0 to 3 volts. Threshold value for 85degrees is taken

	as 750 mill volts.
Battery voltage	0 to 15 volts. Signal conditioned as from 0 to 3 volts.

The output coming from the ADC block is given to another sub-system called the average. Owing to the fluctuations in the input voltage, this averaging technique becomes an essential part in any software design. To get an accurate result, it is necessary to include this block. When expanded, it can be seen as below. It can be observed that 4 values of the input are continuously sampled and averaged to obtain an accurate value void of fluctuation.

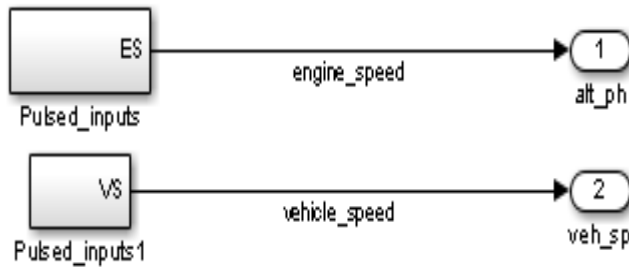


The rate transmission block shown above is included in order to match the rate of data flow from one block to another.

Following this average sub-system is the gain block. This block converts the digital value back to the original value in milli volts in order to ease the programming logic. This averaging technique is implemented for both the analog inputs.

9.1.3 PULSED INPUTS:

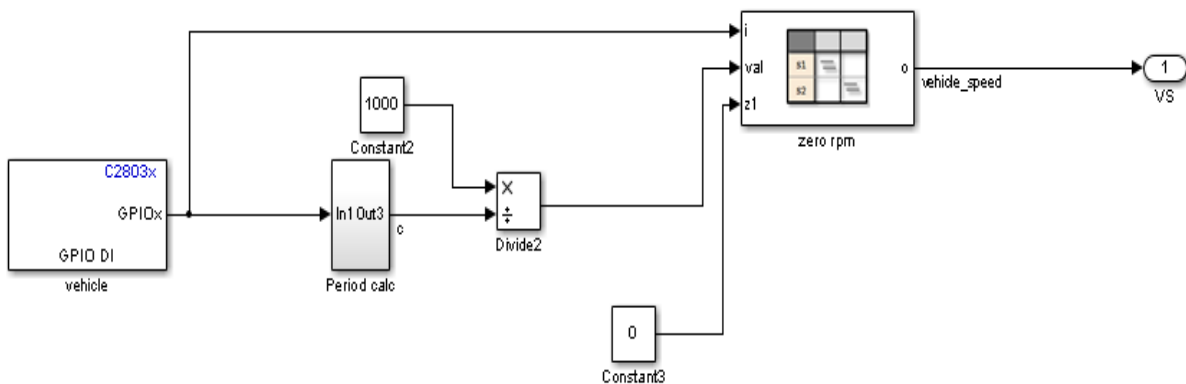
This block contains the analog inputs required for the system. When expanded,



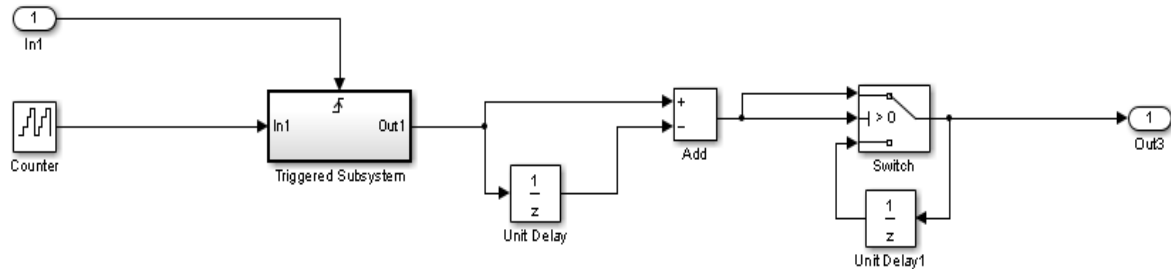
It has two pulsed inputs. One is the engine speed and another is the vehicle speed. Both the inputs are signal conditioned to be within the 3.3 voltage range. The frequency consideration for the two inputs is as follows.

SIGNAL	FREQUENCY CONSIDERATION
Engine speed	160 Hz corresponds to 850 rpm 240 Hz corresponds to 1200 rpm
Vehicle speed	12 Hz corresponds to 6 kmph

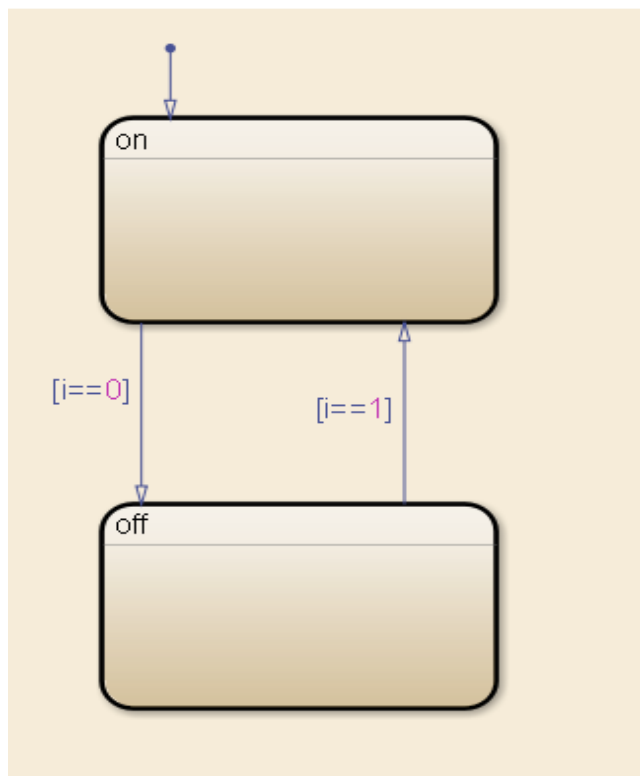
When expanding the vehicle speed block,



A general purpose digital pin is taken as input. It is given to the period calculation block. When expanding the period calculation block,



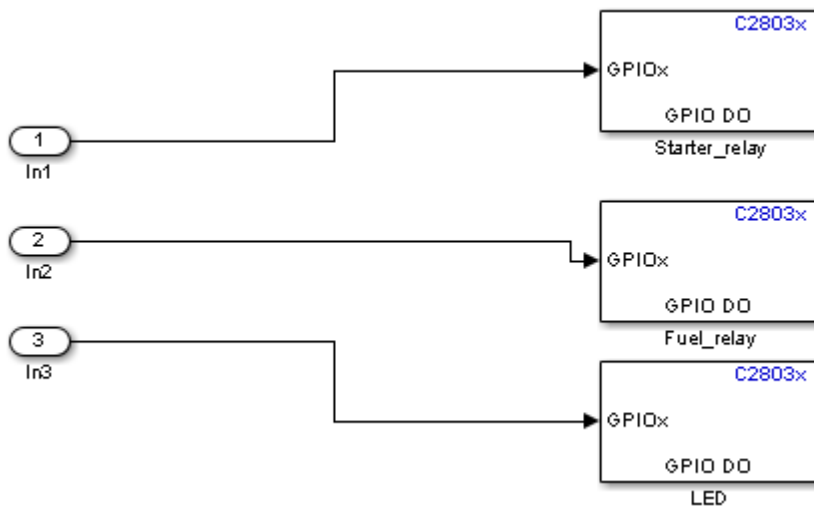
A counter is run at the background. Whenever the input logic level changes, the trigger block is executed. This trigger block outputs the counter value at that particular instant. Subtracting this counter value from the previous counter value gives the period of the incoming pulse signal. The period is divided to obtain the corresponding frequency. The zero rpm block is a state transition table expanded as below.



In order to avoid spurious result when the input pulse is absent, this kind of logic is executed. This block constantly polls the input signal in order to ensure the presence or absence of it. For example, if the minimum frequency of the given signal is fixed as 12 Hz, and if the logic does not see any transition within 83 ms, then it declares the signal to be at zero rpm.

9.1.4 OUTPUT:

This block determines the output going out from the controller. In this case, three outputs are used. General purpose digital pins are configured as output pins.



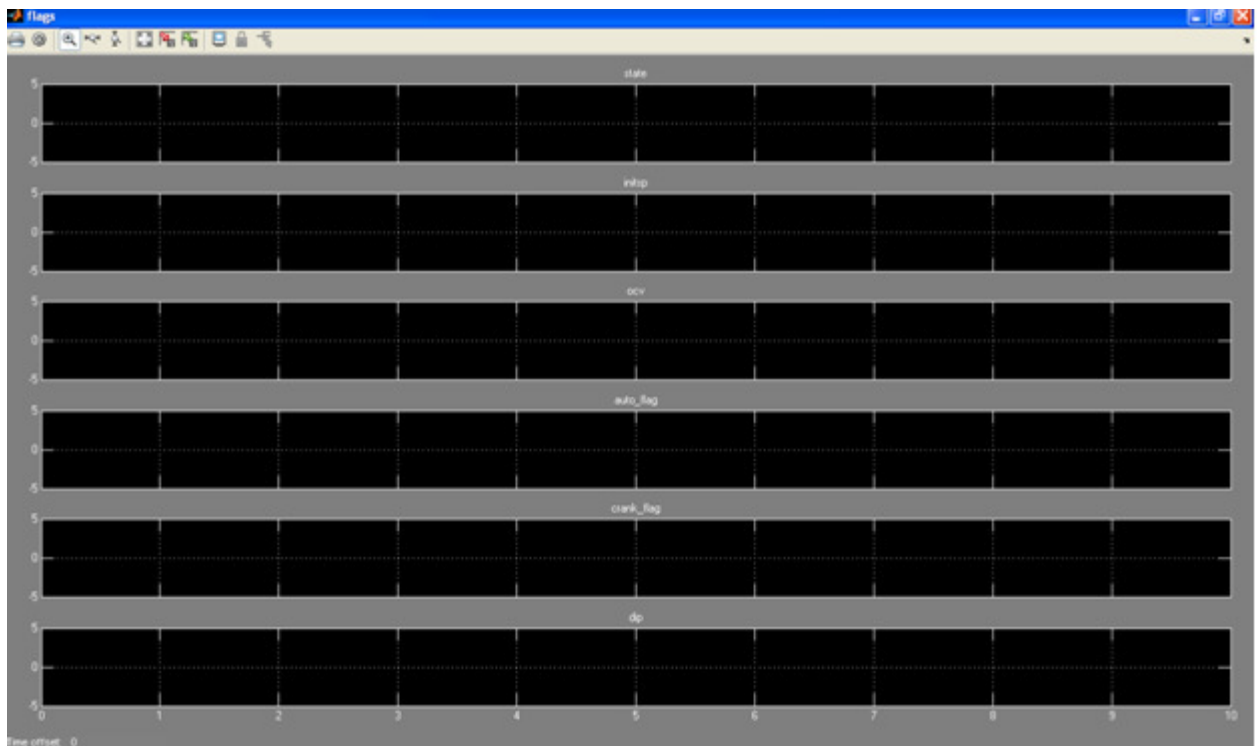
. The following describes the significance of logic levels of the output signals.

SIGNAL	LOGIC HIGH	LOGIC LOW
Starter relay	Turns on the starter.	Turns off the starter
Fuel relay	Cuts the fuel supply	Allows the fuel supply
LED	Auto-stop has occurred. 2 second continuous blink- auto-stop is about to occur. Blink once-when ignition is ON	No auto-stop.

9.1.5 FLAGS:

These are the local variables used for debugging purpose. The flags used are

- State: It is used to identify the state number
- InitSp: it is used to indicate if the vehicle has reached 6 kmph
- OCV: it is used to indicate if the open circuit voltage is within the threshold or not. A logic one indicates it is less than the threshold
- Dip: used to indicate if the battery has dipped or not. A logic one indicates it has dipped.
- Auto-flag: it is used to indicate if auto stop has occurred or not
- Crank-flag: it is used to indicate if engine has fired because of cranking or push-start.



9.1.6 TARGET PREFERENCE:

This block is used to determine the settings of the controller that we are going to use. Here, we are using TMS320F28035 controller. The memory sections can be defined in this block. This block is needed to generate the C code and to link with the IDE (code composer studio).

9.1.7 START-STOP CONTROL- STATEFLOW BLOCK:

This block is where the control of the system is been performed. This block consists of a number of states, the transition between the states and the action that is unique to each state.

The following are the list of states that has been used in the state-flow with a brief description of their purpose.

1. Booting

This is the initial state where the LED blinks once confirming the working of the controller. This state is reached when the controller is turned on through the ignition key.

2. Cranking

This is the state in which the user tries to start the engine. The starter safety measures are taken out in this state. If the engine is fired in this state, the flow transfers to the normal running condition.

3. Key release

This is the state where the user releases the key on account on unsuccessful cranking.

4. Clutch release

This is the state where the clutch releases the key on account on unsuccessful cranking. This state is effective only when the vehicle is started after auto stop.

5. Over cranking

This is the state where the starter has been operated for more than four seconds and is forced to shut down on account of safety measures. This state also indicates the unsuccessful cranking of the engine.

6. Engine off not due to auto-stop

This state indicates that the engine is off initially or engine is off due to reasons other than the auto-atop feature.

7. Engine off due to auto-stop

This state indicates that the engine is off due to the auto-atop feature.

8. Bonnet open – fuel cut

This state is a safety measure feature. When the bonnet is open, the engine is forced to shut down by turning off the fuel supply. This avoids accidents caused due to open bonnet. Care has to be taken while implementing this safety feature.

9. Auto stop – fuel cut

The fuel supply is cut due to the auto stop features. This is indicated by LED on condition. By stopping the fuel supply, the engine is automatically shut down.

10. Engine ON, vehicle not reached the initial speed

This is the initial running condition of any vehicle before it can reach a certain threshold.

11. Engine ON , vehicle above 6 kilometer per hour

This is the state where vehicle runs above the threshold speed.

12. Vehicle crawling after reaching initial speed

This state shows that the vehicle is moving slowly, but has not stopped. This state typically predicts that the vehicle is in traffic signal.

13. Engine idling , vehicle not moving after reaching initial speed

This state indicates that vehicle is not moving and since initially it has crossed the speed threshold, the auto-stop feature can be enabled.

14. Auto stop mode

This state indicates that vehicle is not moving, the auto-stop switch is enabled and other conditions are full-filled and the system is ready to auto-stop.

15. LED blinking

This is an indication to the user that the engine is going to shut down due to auto-stop feature. During this time, the user can disable auto-stop or he can allow the feature to take place.

16. Push start initially

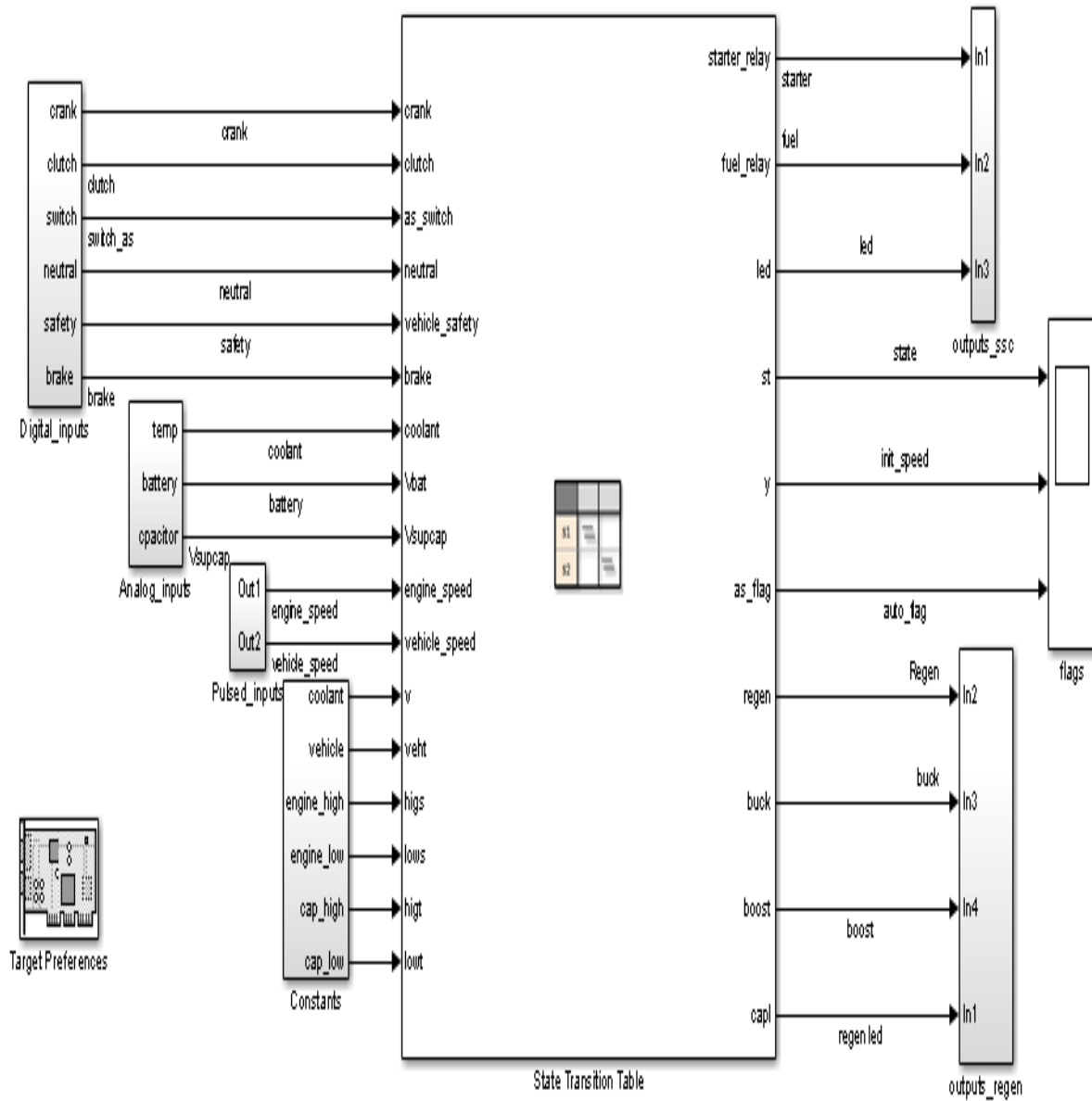
When the battery is in dip condition, the vehicle can be started by pushing it.

17. Push start after auto stop

This state is similar to the previous state occurring after auto-stop condition. This condition is a rare occurrence.

9.2 Regen cum start-stop using ultra-capacitor

Simulink model:



The sub-systems are similar to that of start-stop control. Except for the inclusion of the following signals

SIGNAL	INPUT/ OUTPUT	DESCRIPTION
Brake	Digital input	Logic high – Pressed
Super capacitor voltage	Analog input	Ranges from 0 to 15 volts.
Regen	Digital output	Logic high- Alternator to Super capacitor. Logic low- Alternator to battery
Buck	Digital output	Super capacitor to Battery
Boost	Digital output	Battery to Super capacitor
Regen LED	Digital output	Glowes when Super capacitor is charging

The debounce logic for Brake signal and the averaging technique for the Super capacitor voltage are provided as mentioned previously.

The output block has extra 4 outputs in addition to that of start- stop. It has the similar digital output blocks.

The following are the list of states used with a brief description of their functionality. It contains all the 17 states of start- stop. In addition to these, the following states are incorporated

18. Initial super capacitor charge status

Before the engine can be turned ON, the super-capacitor charge status is checked for a healthy cranking.

19. Super capacitor charging during successful cranking

When the cranking fails due to insufficient charge status, this state is arrived.

20. Super capacitor charging during normal running

The super-capacitor voltage is maintained between the lower and upper thresholds. In order to achieve this during the vehicle running condition, this state has been created.

21. Super capacitor charging after auto-stop

Super-capacitor has to be charged well above the minimum threshold after auto-stop in order to facilitate starting of the engine through clutch start.

22. Braking

When braking happens, the control flow comes to this state.

23. Regeneration

Upon satisfying the regeneration condition, the control flow moves from braking to this state. In this state, the energy recovery occurs.

24. Discharging super-capacitor

This state follows immediately after regeneration in order to transfer the acquired energy to the battery and also to be ready for absorbing energy during the next braking event.

25. Regeneration plus bucking

When brake is pressed continuously and super-capacitor reaches its maximum capacity to absorb the energy, the buck convertor is activated in this state.

In addition to these states there can be funny scenarios in which, we cannot determine the status of the program flow. For example, suppose that the controller is normally working, but it is suddenly reset by some external disturbance. During this kind of a situation, it is unlikely that a programmer can give assurance to the validity of his code. This is where Stateflow comes handy. We can define a state called the “ambiguous” state or an “undefined” state which is responsible to solve ambiguities of this kind. This state will be responsible to sense the input status and map it to any of the state in the state-flow.

9.3 Code generation:

The above mentioned blocks have been properly set with appropriate sample times for each block. Sample time is the rate at which the output is updated. The code is generated by building it from the Simulink and linking it to the IDE code composer studio. The generated code can be modified to include the programming lines to support flash programming. Also, the interrupt routines, if needed, have to be written manually and built again. Finally, the code is burnt into the flash memory. The code can be run under debug mode to watch the signals in CCS watch window or it can be run as an independent target.

CHAPTER 10

RESULTS AND DISCUSSION

The generated code was dumped into the target and the results were obtained as expected. The following shows the different test conditions. The signals are captured in scope for better understanding and validation.

Case 1:

Ignition is in OFF condition. The controller is not powered now.



Case 2:

Ignition OFF to ON condition. LED blinks once ensuring that the ECU works properly.

At this time, the controller is in BOOTING state.



CASE 3:

Successful cranking occurs. Engine is fired. Vehicle safety is maintained to be in high logic ensuring that the bonnet is closed.



CASE 4:

Auto-stop condition: When vehicle speed approaches zero, provided the auto-stop conditions are satisfied, there is a two second delay followed by LED blinking. Later, the fuel is cut to shut down the engine. LED is continuously ON indicating that engine is shut due to auto-stop feature.



CASE 5:

Clutch start after auto-stop. The engine starts after auto-stop by pressing clutch. The vehicle moves. Upon satisfying auto-stop conditions, the engine is again shut. This is indicated by the vehicle speed becoming zero and by the blinking of LED.



CASE 6:

The following shows the hindrance during auto-stop .The user can decide against allowing auto-stop to occur by either pressing clutch or by disabling the auto-stop enable disable switch or by changing the gear position.



CASE 7:

Over- cranking. In order ensure the safety of the starter; the starter is not continuously operated. A delay of 2 seconds is given between successive starters ON condition. The starter can be ON for a maximum of 4 seconds.



CASE 8:

Even when vehicle is pushed and started without starting through the key, auto-stop feature is enabled.



CASE 9:

When the battery dip has occurred, auto-stop does not occur. But, in case of Regen cum start-stop, the battery is assumed to be fully charged.



CASE 10:

If the vehicle speed is zero, before the vehicle could attain a certain speed, auto-stop feature is not enabled. Only when the vehicle becomes zero after attaining the initial speed of 6 kilometer per hour, this feature is enabled. In the following, it can be seen that when the vehicle speed is less and it becomes zero, LED does not glow. But, when the speed is increased and reduced to zero, auto-stop occurs.



CASE 11:

Initial super-capacitor charging: this is indicated by the LED_CAP and the boost signal.

The charging stops only after the capacitor reaches a maximum threshold level.



CASE 12:

Initial super-capacitor charging: this stops as soon as the crank signal is received and the capacitor voltage is well above the minimum threshold limit. The logic does not wait for the maximum threshold to be reached. Priority is given to the crank signal.



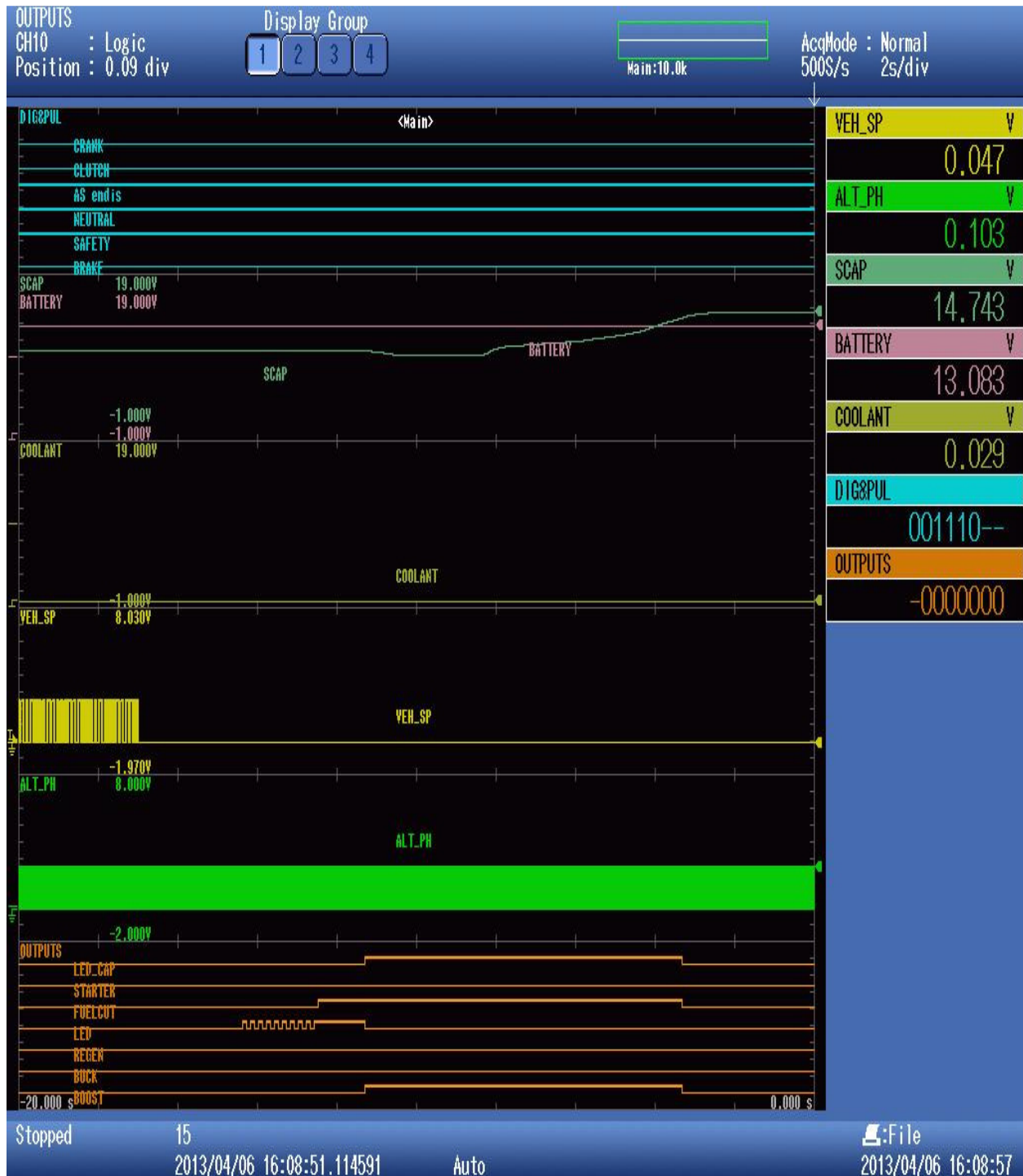
CASE 13:

Super capacitor charging during normal conditions: this is to ensure that the capacitor is maintained between the upper and lower thresholds. This validates the balance between the ability of the super-capacitor to start the engine as well as to absorb the kinetic energy during braking.



CASE 14:

Super capacitor charging after auto-stop: this ensures that engine can be started without appreciable delay. Here, auto-stop is indicated by the blinking of LED and the super-capacitor charging is indicated by the LED_CAP and boost signal.



CASE 15:

This shows that even when brake is pressed and clutch is not pressed, regeneration happens only when the vehicle speed of the auto mobile has reached above a certain value. This can be viewed by the increasing frequency of vehicle speed.



CASE 16:

This shows that even when brake is pressed and vehicle speed is above certain value, regeneration will not occur since the clutch is pressed. When the clutch is released, regeneration will occur.



CASE 17:

This shows the normal running vehicle. Braking event takes place. Clutch is not pressed. Regeneration happens. Brake is released. Super capacitor is discharged by activating the buck convertor. When it has reached the lower threshold, the discharging stops.



CASE 18:

This condition shows that during braking, regeneration happens. When the super-capacitor reaches the maximum threshold value, the buck convertor is established between super capacitor and battery. This condition enables an increased level of energy recovery.



CASE 19:

Repeated key start: this shows that when the key is continuously engaged to start the vehicle, the signal to the starter goes only after a 2 seconds delay. This ensures the protection of the starter and increases its life span. This safety precaution is suited even during clutch start after auto-stop.



These are some of the test patterns with verified results. These results have also been obtained in the Simulink environment and found to be true.

Thus, the software logic for the start-stop control and Regen cum start-stop control has been developed and tested at simulation level. The code generated from this Simulink model has been dumped into the hardware circuitry and the above stated results were obtained.

The scope used here is the Yokogawa DL850 Scope Corder. All the input and output signals have been tested and captured successfully at the laboratory level.